# Advanced Educational Activities Pvt. Ltd.

## (Key and Solutions for AIMCAT2413)

## Key

### SECTION – I

1. C
2. B
3. A
4. D
5. A
6. C
7. C
8. B
9. A
10. B
11. D
12. A
13. A
14. D
15. A
16. B
17. B
18. B
19. 3241
20. D
21. C
22. B
23. 3421
24. 4213

### SECTION – II

1. B
2. 170
3. D
4. C
5. A
6. 385
7. A
8. D
9. C
10. 50
11. B
12. C
13. D
14. B
15. A
16. 36
17. 10
18. B
19. C
20. 78

### SECTION – III

1. D
2. 0
3. A
4. 26
5. B
6. C
7. 24
8. D
9. B
10. B
11. C
12. 17
13. 540
14. 2
15. 11
16. D
17. A
18. D
19. D
20. A
21. B
22. 51

## Solutions

### SECTION – I

**Solutions for questions 1 to 4:**

**Number of words and Explanatory notes for RC:**

Number of words: 515

**1.** Option A: The philosophy of the Middle Age was supposed to be a low according to some philosophers, and the author disagrees with that saying the philosophy of the Middle Ages was sophisticated as well. However, we cannot infer from either of those options that the philosophy was the same as that of the subsequent eras. Option A is not the answer.
Option B: The author strongly argues against such a conclusion by saying that the Middle Ages were referred to as Dark Ages because of a lack of knowledge or clarity. The Dark Age in fact were richly philosophical according to the author. Option B is not the answer.
Option C: The author believes that ***'It (Middle Age) is instead extraordinarily rich philosophically, and should be celebrated as hugely innovative.'*** From this, we could infer two things, that the philosophy of the Middle Age was advanced/modern and that it was under appreciated. Option C is the answer.
Option D: The period referred to by Brucker as Dark Age/Middle Ages precedes the 18th century because that was when Brucker published his work. Hence, Option D is not the answer. Choice (C)

**2.** Consider this: ***'In his book The Classics and Renaissance Thought (1955), he notes that Renaissance humanism is 'a broad cultural and literary movement, which in its substance was not philosophical, but had important philosophical implications and consequences.'*** The author is talking about the book to introduce the idea that humanism is a cultural and literary movement with only philosophical implications.
Option A: The argument is not whether the use of the term 'renaissance' is justified or not. It is about the mention of the book, the purpose of which was to talk about what Renaissance humanism was actually about. Option A is not the answer.

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Option B: This is the purpose of mentioning the book, to explain that the term Renaissance philosophy is more associated with culture than with philosophy and hence is wrongly attributed. The author goes on to explain that there never really was a philosophical movement, per se. Option B is the answer.
Option C: While this is partially true, the purpose of mentioning the book had a different tone to this particular option. The book was mentioned not to highlight that Renaissance humanism had some philosophical implications. It was more to highlight that it was more associated with culture than with philosophy despite the philosophical association. Option C is not the answer.
Option D: The example of the book was not to discuss coherence of the use of these words 'in philosophy'. The discussion is about the association of the concepts (rather than the terms) Renaissance and humanism with culture more than philosophy. It's not completely distinct from philosophical connotations, yes, but it is predominantly a cultural thing. But whether these words can be used 'in philosophy' is not at all the discussion here (the passage does not discuss the content of philosophy here). Option D is not the answer. Choice (B)

**3.** Option A: The author clearly states this about the terms renaissance and humanism, that ***'They were conceptions of certain perceived historical developments or movements in art, and not formed to fit philosophy.'*** So, the author will approve of the use of these words to depict an era of art different from the previous forms. This can further be understood from ***'... also made use of 'humanism' and 'Renaissance' as a means to discuss a particular era in the history of art. Both terms were formed in the tradition of Brucker and were intended to capture something new.'*** Hence, Option A is the answer.
Option B: The author strongly argues against the idea that philosophy went through any form of decline during the Dark Ages and resurfaced out of it as Renaissance philosophy. So, the author will disapprove of this statement unequivocally. Option B is not the answer.
Option C: The author strongly disagrees with the idea that the Middle Age was some kind of a Dark Age of ignorance. In fact, it is referred to as a period of innovation in philosophy by the author. So, the author wouldn't associate the word Renaissance with 'art resurfacing as a new movement', even though Renaissance is, in fact, associated with a movement in art. Option C is not the answer.
Option D: The author tries to explain that there was no philosophical movement per se called Renaissance philosophy. It was more an art and culture movement. So, there is nothing to draw a parallel between. Option D is not the answer. Choice (A)

**4.** Option A: The author talks about how humanist principles in philosophy weren't new, that they were part of ancient philosophy as well. The author doesn't say the same about movements or principles in art. Hence, Option A is not the answer.
Option B: The author argues that there was never a separate movement in philosophy, even though historians tried to create a term for it called Renaissance Philosophy. So, the idea that such a movement didn't succeed is wrong. Option B is not the answer.
Option C: The author clearly argues in the passage that there was no decline in philosophy, per se, and that the Middle Ages were equally innovative. So, the author wasn't trying to highlight the decline, just argue that there was nothing called a renaissance movement in philosophy. Option C is not the answer.
Option D: This comes closest to explaining what the author was trying to do in the passage: discuss and reason about whether there really was a movement in philosophy. This can be understood from the first para itself when the author says that ***'This is a deeply problematic characterisation, partly based on the assumption of a conflict between two philosophical traditions – a conflict that never actually existed.'*** Option D is the answer. Choice (D)

**Solutions for questions 5 to 8:**

**Number of words and Explanatory notes for RC:**

Number of words: 527

**5.** Option A: This is the reason the author draw a parallel as understood from the statement: ***'Indeed, absolutist [monarchy was] limited in scope by the strong legal traditions that prevailed across Europe,'*** and ***'But because the unaccountable sovereign ruled through an increasingly institutionalized bureaucracy, there was regularity and transparency in the government's behavior that over time evolved into a legal constraint on arbitrary despotism.'*** Option A is the answer.
Option B: The Prussian state had a legal tradition based in bureaucracy, whereas the Christian Europe had rule of law based in religion. Hence Option B is not the answer.
Option C: The 11th century Christian Europe and 19th century Prussian State did not hinder the functioning of their monarchies. They restricted the power that the monarchy can wield – the former with the help of religion and the latter with the help of bureaucracy. This restriction of power may not necessarily have hindered how the state (or monarchy) functions. Hence, Option C is not the answer.
Option D: Neither did the Prussian state nor did the Christian Europe have an unaccountable sovereign particularly because they were accountable, at least partially, to a legal higher order, bureaucracy in one case and religion in another case. Hence, Option D is not the answer. Choice (A)

**6.** The assumption underlying the secular grounding has been explained here: ***'the justification for this, according to these new theorists, lay in the fact that they in some sense "represented" the broader interests of the whole community.'***
Option A: This explains what the secular grounding of absolutism means. It doesn't explain what the assumption underlying this power of monarchs is. Hence, Option A is not the answer.
Option B: This is not the precondition or prerequisite, this is the outcome. Sovereigns have absolute authority free of religious legal constraints. The precondition for that absolute authority is that monarchs represent the broader interests of the community beyond the concept of a god/religion (hence, secular). Option B is not the answer.
Option C: This represents the underlying assumption. As understood from the words in bold above, monarchs managed to wriggle out of religious authority and claim absolute authority with the sense that they represented the broader interests of the whole community. The assumption was that the interests of the community come first, ahead of any religious authority, and that was the monarchs' duty to foresee. Option C is the answer.
Option D: Sovereigns had to represent the community: this is the assumption. However, there is no comparison mentioned in the passage, that they had to represent the community more than they represented the religion. In fact, the secular grounding means that they didn't have to represent the religion. Option D is not the answer. Choice (C)

**7.** Option A: Absolute monarchs were not necessarily beyond the constraints of Prussian law since some of them (the strong-minded ones) defied the law. For them to defy the law, they should have been bound by the constraints of the law in the first place. Also, not all of the absolute monarchs defied the law. Hence, this is not the answer.
Option B: The ordinary citizens benefited from the law. However, the rulers were also bound by the same law, which some of them defied. Hence, option B is not the answer.
Option C: The author mentions that strong minded rulers can defy the law but the citizens were bound by it. This can be inferred from ***"And yet ordinary citizens in their dealings with one another and with the state could expect increasingly uniform and impersonal treatment"***. Hence, option C is the answer.
Option D: The passage mentions that strong-minded rulers can defy the law but does not provide any information on the situations in which they defy the law. This option limits the defiance to when the ruler's authority was questioned. There is no mention of ruler's authority being questioned in the passage. Hence, this is not the answer. Choice (C)

**8.** Option A: Princely power was now constrained by a social contract (to guard the fundamental right to life of every citizen). However, it is not correct to say that 'it lost its agency' that it enjoyed earlier. It gained more agency, in fact, because the absolute rulers were now free of the need to be a representative of divine sovereignty. Option A is not the answer.
Option B: The author says that despite the secular grounding of absolutism in the state the princely power was not absolute. The rulers were now free from the need to be constrained by religious authority where the king was just an agent for the power of God rather than being the supreme voice of the state. However, the rulers were still not able to take absolute power for themselves because their new-found power to go beyond the religious constraints was based on the understanding or implicit social contract that they will work for the broader interests of the community. Option B is the answer.
Option C: Princely power was undermined by the need to fulfil the implicit social contract to guard the best interests of the community and, by extension, all citizens. The author doesn't say that it was undermined specifically by transparency. Option C is not the answer.
Option D: Princely power was not absolute because their agency was rooted in a certain expectation - that they will guard the interests of the broader community and hence the 'fundamental right to life' of citizens. This is not the same as saying their power was undermined because of their duty to guard 'the fundamental rights' of the citizens. We are not even sure if citizens had any fundamental rights, per se. Option D is not the answer. Choice (B)

**Solutions for questions 9 to 12:**

**Number of words and Explanatory notes for RC:**

Number of words: 522

**9.** Consider this statement: ***'The question, which we may never have the evidence to answer, is whether it is just an accident that humans descended from a chimp-like ancestor rather than a bonobo-like one, for it may well have been that the aggression and violence of the chimp-like progenitor of both humans and present-day chimps was what drove intelligence, sociability, and a host of other cooperative characteristics of human nature.'*** So, the author believes we don't have the answer to whether humans descending from an aggressive chimp-like ancestor is logical or accidental.
Option A: The aggression of our chimp-like progenitor drove human intelligence, sociability and cooperative characteristics. So, it is true that it is this aggression of the chimp-like ancestor that shaped the direction of human evolution, which the author establishes is based on cooperation and cognition. Hence, Option A is the answer.

Option B: While the author mentions a host of cooperative characteristics, there is no direct mention of the connection between the aggressive ancestor and the competitiveness of humans as understood from the lines quoted above (in bold). Hence, Option B is not the answer.
Option C: The author tries to analyse whether humans descending from chimp-like ancestors is accidental or by design. So, it is clear that humans and present-day chimps share an ancestor. But it is equally clear that humans and present-day chimps don't share an ancestor with bonobos which are less aggressive and more happy-go-lucky creatures. Option C is not the answer.
Option D: The author doesn't analyse whether humans and present-day chimps evolved differently from each other. The analysis is about how humans and present-day chimps have evolved differently from bonobos, and whether that is accidental or by design. Hence, Option D is not the answer. Choice (A)

**10.** The argument made by Humphrey and Alexander in the last paragraph of the passage can be understood here: ***'the most important and dangerous part of a human being's environment quickly became other human beings, such that developing cognitive skills for social interaction quickly became the most critical requirement for evolutionary fitness...'*** They argue that human brains evolved as a direct consequence of the need for improving their social interactions in order to deal with their biggest danger which was other human beings.
Option A: If there are other species where members are dangerous to each other, it contradicts the argument that human brains evolved to deal with the danger of their environment caused by the members of their own species. We cannot say that humans had the necessity to grow bigger brains because other humans were dangerous (since it is not seen in other animals facing the same danger). Option A is not the answer.
Option B: Even if human culture is proportionally less evolved per unit size of the brain compared to other animals, this option doesn't weaken the argument since Humphrey and Alexander don't base their entire argument on the complexity of the human culture. Further, we cannot infer whether cooperation leads to a complicated culture. Cognitive intelligence may involve other parameters outside the ability to create a culture. So, their argument that the need for social interaction and cognitive skills led to the brain size increase can still hold true. Option B is the answer.
Option C: Their argument is that human brains evolved and grew in size to cope with the demand for cognitive skills needed for social interaction. If intelligence is not dependent on the size of the brain, then this argument falls apart. It means the brains growing in size cannot be correlated to the need for cognitive skills. Hence, Option C is not the answer.
Option D: If true, this statement shows that brain size and cognitive skills are not directly influencing each other. That means the growth in brain size in humans need not be credited for their increased cognitive skills and vice-versa. So, this option weakens the argument. Option D is not the answer. Choice (B)

**11.** Of the four options, three options, if true, will agree with the passage, thus strengthening the arguments. We are looking for an option that doesn't strengthen the arguments in the passage.
Option A: This, if true, will agree with the arguments presented in the passage. The rapid growth of the human brain has been because of the need to compete in their dangerous environment, the danger being other humans. Hence, Option A is not the answer.
Option B: If true, this agrees with the passage. The author says that defensive modernisation happens because of foreign threat, leading to internal political changes as well, to facilitate a literal arms race. Hence, Option B is not the answer.
Option C: The passage depicts that humans cooperated in order to compete with other human groups. Competition and cooperation both drove evolution. Cooperation was needed as groups formed and competed with each other. The ability to cooperate provided an advantage to humans as far as competition is concerned. If true, this option agrees with the passage. Hence, Option C is not the answer.
Option D: Even if true, this doesn't strengthen the arguments in the passage, as the author explains that both cooperation and competition were parallel qualities that were needed for the human evolution. So, it is not that one outpaced the other, or one quality was present in a greater measure compared to the other. Hence, Option D is the answer. Choice (D)

**12.** If it is proven that humans cooperate even when it doesn't give them a competitive/evolutionary advantage, it contradicts the argument that cooperation is to survive in competition.
Option A: If true, this option shows that humans cooperate even when there is no competitive advantage. This contradicts the argument made by Alexander that cooperation has an agenda, i.e., to compete with other groups. Option A is the answer.
Option B: This option talks about when humans compete but doesn't really bring cooperation into the picture. From here, we cannot understand how and whether cooperation and competition are interrelated. Option B is not the answer.
Option C: This option talks about the importance of competition for humans in order to survive. It is not relevant to the question about the correlation between competition and cooperation. Hence, Option C is not the answer.

Option D: Even in the face of impossible odds (great difficulties stacked against them), humans cooperate. This option can be made to support the argument that cooperation is necessary to compete. So, it won't contradict the words that human beings cooperate to compete. Option D is not the answer. Choice (A)

**Solutions for questions 13 to 16:**

**Number of words and Explanatory notes for RC:**

Number of words: 515

**13.** Consider the author's remarks against the populist argument: ***'Current shareholders who sell their shares do so voluntarily. The term takeover does not seem compatible with the voluntary sale.'*** So, we are looking for an option that contradicts the concept of voluntary sale of shares by shareholders.
Option A: The author assumes that the sale of shares will leave the shareholders richer. This option provides a scenario in which shareholders do not sell their shares at high prices as the authors seems to assume. Hence, this option is incompatible with the author's remarks against the populist argument. Hence, this is the correct answer.
Option B: Even if this were true, at the end of it shareholders do sell away their shares voluntarily. Hence, this option is not incompatible with the author's remarks. Option B is not the answer.
Option C: Even if this were true, this is an issue with the price and not with whether the sale is voluntary or not. Hence, it is irrelevant to the argument made by the author. Option C is not the answer.
Option D: Even if this option were true, it deals with what happens to the company after the takeover and what the so-called 'raiders' do with it. This option doesn't have a bearing on the argument here about whether the sale of shares is voluntary. Option D is not the answer. Choice (A)

**14.** The assumption is something that has to be true for the author's conclusion to be true. The conclusion is that the news of the takeover drives up the share prices of the firm.
Option A: The share prices fluctuating is neither good nor bad, as fluctuating means both up and down. Secondly, this option doesn't talk about whether the fluctuation and probabilities are correlated in any particular direction. Hence, this is not the basis for the author's argument. Option A is not the answer.
Option B: The author does not base his argument on the assumption regarding whether the investors are capable or incapable. That is irrelevant to the discussion. The share price increases because the news of a takeover indicates a greater confidence in the earning capability of the company. Hence, option B is not the answer.
Option C: Earning potential of the company relates to a different discussion. The author's argument is about the correlation between share prices and the news of takeover, which has got nothing to do with the firm's earning potential. Option C is not the answer.
Option D: If a group of investors is trying to takeover the company, it indicates that the company can become more profitable in the future (*'by replacing the firm's current managers with new managers, the raiders make the firm more profitable'*). Better future prospects result in increased share prices before a takeover. Hence, Option D has to be true for the author's conclusion to be true, and therefore, this is the underlying assumption. Option D is the answer. Choice (D)

**15.** Option A: This is not true. The shareholders are the actual owners of the corporation and on their behalf, it is the managers who have the agency. However, during a takeover, such agency can be taken away and handed over to a different set of managers. Option A is the answer.
Option B: This is an inference that can be made from the statement that ***'Sometimes a group of investors believes that a different team of managers can operate the firm more profitably.'*** So, shareholders can also use their agency to choose which set of managers can call the shots at the corporation based on their potential. Hence, Option B is not the answer.
Option C: This too can be understood from the statement ***'The third major area in which economists have contributed to the analysis of corporate takeovers is the empirical determination of which parties gain and which parties lose in contests for the control of corporations.'*** So, the studies can help us decide who gains wealth and who loses in the process of corporate takeovers. Option C is not the answer.
Option D: This can be understood from ***'Using event studies, economists have established that the shareholders of corporations that are taken over almost invariably gain wealth.'*** Takeover is the equivalent of competition between alternative teams of managers. Hence, Option D is not the answer. Choice (A)

**16.** The invalidation the author talks about is that the populist argument that the raiders (those who are taking over the corporation) do not act in the interests of the shareholders is incorrect. Supporting the invalidated argument would be to prove that the raiders indeed harm the interests of the shareholders.
Option A: How the firm does after the takeover is outside the purview of the discussion as the shareholders interests are already benefited through the sale of the shares and gaining of wealth. The investors who purchases shares from the shareholders might lose but that doesn't support the invalidated argument. Option A is not the answer.

Option B: The invalidation mentioned by the author is based on the argument that shareholders gain wealth during a takeover. So, it invalidates the populist argument that shareholders lose because of a takeover, or that shareholders' interests are harmed by those who effect the takeover. However, if the shareholders lose wealth, their interests are indeed harmed by the takeover. This supports the invalidated argument. Option B is the answer.
Option C: Sentimental reasons are outside the purview of the discussion which is around whether the shareholders gain or lose monetarily because of the sale of shares during a takeover. Option C is not the answer.
Option D: What plans the raiders have for the future of the company is not relevant to the discussion here about whether the shareholders gain or lose from the takeover. Once the shareholders gain wealth and the takeover is complete, it doesn't matter what the raiders do with the firm. Hence, Option D is not the answer. Choice (B)

**Solution for question 17:**

**17.** Option A: Longer deadlines are not counter-productive just because they induce lead consumers to "waste" resources. Committing more resources is only one of the side-effects but the difficulty inference and wrong perceptions are what cause the sabotaging of the goal pursuit. Option A is not the answer.
Option B: This option provides the best possible essence of the paragraph: that longer deadlines are detrimental to goal pursuit and they are detrimental because they change the way lead consumers look at difficulty of the task. They think that the goal is more difficult than it actually is, and this perception sabotages the process. Option B is the answer.
Option C: Longer deadlines don't make goal pursuit more difficult to achieve because of the commitment to unnecessary resources. This is the wrong causation. They sabotage goal pursuit because of a perception difference, making the goal seem more difficult (not actually make it more difficult). Option C is not the answer.
Option D: The paragraph talks about lead consumers making wrong inferences about the difficulty of the goal. It doesn't talk about unintended difficulties arising out of those wrong inferences. It talks about unintended difficulty perception, not unintended difficulties. Option D is not the answer. Choice (B)

**Solution for question 18:**

**18.** The highlighted sentence mentions friendship, which is not referred to in the statements immediately preceding and following blank (1). Therefore, inserting the highlighted sentence between them would break the thought flow. Hence, the highlighted sentence does not fit into blank (1).
The statement immediately preceding blank (2) talks about friendship being an act of recognition and belonging. The highlighted sentence extends this idea by stating that this act of belonging cuts across all convention, morality and category. The statements following blank (2) elaborate on the cutting across categories like space or time on the soul. Therefore, these statements substantiate what is given in the highlighted sentence. Hence, the highlighted sentence fits into blank (2).
The idea of anam cara and a friendship cutting across all constraints, is upstream to the idea that you were joined in an ancient and eternal way, which is more like a conclusion from the statement in the blank (2). After this we move on to talking about the Celtic tradition which is not compatible with friendship cutting across conventions. If we place the bold statement in (3), we are going from 'ancient and eternal' to 'cutting across conventions' to 'Celtic tradition not setting limitations, which doesn't have any logical flow to it. Hence, (3) is not the right blank for the bold statement.
The statements immediately preceding and following blank (4) talk about limitations on the soul. The highlighted sentence talks about how "your friendship" cut across all convention, morality and category. Inserting the highlighted sentence in blank (4) would break the thought flow of the statements immediately preceding and following it. Hence, the highlighted sentence does not fit into blank (4). Choice (B)

**Solution for question 19:**

**19.** Sentence 1 is referring to 'this' adaptation of the language and talks about how English that was distinctively African was developed. So, 1 will follow the idea discussed in 2, of which 4 is an example.
Sentence 2 discusses the transcreation of African cultures and how African writers went about it to 'adapt and indigenise certain aspects of the language.'
Sentence 3 gives context to 2, by talking about or introducing AfrE, that is further elaborated in 2. So, 3 is a bigger idea compared to 2, and hence precedes it.
Sentence 4, an example sentence, has the pronoun 'such' referring to the 'creative indigenisation' in 2. So, 24 is a mandatory pair. So, 241 has to be a block, since 1 cannot split 2 and 4. So, the order is 3241. Ans: (3241)

**Solutions for questions 20 and 21:**

**20.** Option A: While this option talks about one of the important ideas that Euripides wasn't appreciated by critics because his work lacked dramatic intensity, it doesn't highlight that we are talking about two different eras. The critics are from Euripides times, while Euripides was popular with the modern audiences. Option A is not the answer.

Option B: Euripides' work didn't necessarily gain popularity amongst 'modern critics'. The paragraph simply says he was more popular in the modern times compared to his contemporaries. Modern critics have not been mentioned. Option B is not the answer.
Option C: The author was more unbiased towards Euripides. This option makes it sound like the author favoured Euripides and thought that he deserved recognition. But that's not what the para says. So, the use of the term 'got its due' is incorrect. Option C is not the answer.
Option D: This option is apt since it highlights the main ideas, that Euripides was not acclaimed during his times for those parameters (themes, rhetoric, poetic oratory) which now make him popular. Option D is the answer. Choice (D)

**21.** Option A: While this option is partially true, it doesn't explain the important idea that we are talking about social constructivism from the perspective of world politics. Hence, Option A is not the answer.
Option B: Social constructivism disputes the idea that the social world is external to the people who live in it. It is not the assumption. It is disputed by the theory. Hence, Option B is not the answer.
Option C: Social constructivism has evolved out of certain world events. This is true. The events show that human agency indeed matters. This is also true. Theoretical works support this view and form the basis of social constructivism. This is also true. Option C is the answer.
Option D: This option dispassionately differentiates between realism and liberalism on one side and social constructivism on the other. However, the para is about social constructivism more than liberalism and realism. Option D is not the answer. Choice (C)

**Solution for question 22:**

**22.** The statement immediately preceding blank (1) mentions the EIC rejecting a proposal. The statement immediately following blank (1) states that the EIC "likewise rejected" another proposal. The highlighted sentence, which talks about steps towards industrial development will break the flow if inserted between the two statements. Hence, the highlighted sentence does not fit into blank (1).
The statement immediately following blank (2) mentions that the EIC "also risked" something. This statement should be preceded by a statement that mentions that the EIC risked something. This statement cannot be the statement immediately preceding blank (2) because it talks about the EIC rejecting something. Therefore, the statement immediately following blank (2) has to be preceded by the highlighted sentence. The highlighted sentence does not directly mention any risk. But it can be inferred from the statement "making the colony (India) still more independent of the Mother Country (Britain)" that the risk is India becoming more independent from the British. Hence, the highlighted sentence fits into blank (2).
The statement immediately preceding blank (3) mentions an additional risk ("also risked"). The highlighted sentence mentions one risk. There is no other statement in the paragraph that mentions a risk. Hence, the highlighted sentence should precede the statement immediately preceding blank (3) and not follow it. Hence, the highlighted sentence does not fit into blank (3).
The statements immediately preceding and following blank (4) are specific to Bengal while the highlighted sentence is more about India's independence with respect to Britain. Therefore, these two statements are connected and inserting the highlighted sentence between them would break the flow. Hence, the highlighted sentence does not fit into blank (4). Choice (B)

**Solutions for questions 23 and 24:**

**23.** Sentence 1 is a grammatically independent sentence that discusses the relation between the organisational and technical forces of production change and the social relations of production change.
Sentence 2 starts with a positive connector and refers to a previous sentence as 'all of that'.
Sentence 3 is an independent sentence that discusses the process of social change and what it comprises of.
Sentence 4 refers to some other argument with the connector 'for' used in the meaning of 'because' and explains that society is organised into class structures. This is the cause for whatever sentence precedes 4.
We can understand that 2 has to follow the combination of 3 and 4 which together discuss social change and class structures. 1 elaborates 2 by explaining how social relations of production change with organisational and technical forces. There is another way of looking at this paragraph. 3 and 4 talk about past—class and social structures. 2 and 1 talk about how that changes. Ans: (3421)

**24.** Sentence 1 is contextually dependent because of the usage of the term 'so small and simple' that talks about Brenner's requirement for his gene study.
Sentence 2 explains why the fly was too big for Brenner.
Sentence 3 talks about a cell number as a comparison, which means that 3 will definitely come after 1. That is because it is in 1 that we talk about cell counting, and that is why we need a point of comparison.

Sentence 4 talks about the experiment, year and the person. The full name Sydney Brenner would obviously precede any use of his name. So, 4 is the first sentence of the paragraph.
4 is the intro and 13 is a mandatory block. We cannot split 13 to place 2 because that will make 3 confusing and context-less. And 2 cannot come in the end since the topic discussion has moved on to the process of counting cells. So, the right order is 4213. Ans: (4213)

| *Difficulty level wise summary - Section I* | |
|---|---|
| Level of Difficulty | Questions |
| *Very Easy* | – |
| *Easy* | – |
| *Medium* | 20 |
| *Difficult* | 1, 3, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 21, 23, 24 |
| *Very Difficult* | 2, 4, 18, 19, 22 |

### SECTION – II

**Solutions for questions 1 to 5:**

The number of kids in the eight cities, in the order in which they are mentioned, are 20, 10, 20, 10, 30, 10, 20 and 30. It is given that the total number kids and adults are the same across the eight cities. Let us consider the number of adults in Kantpur to be a (since Kantpur is one of the cities that has the highest population of Kids, the population of adults in every other city will be more than a). Then the number of adults in Antarpur, Blasepur, Chastipur, Dustypur, Luckthen, Herebad and Matpur will be 10 + a, 20 + a, 10 + a, 20 + a, a, 20 + a and 10 + a, respectively.

It is also given that the total number of teenagers, adults and elderlies is also the same across the eight cities.

The total number of teenagers and adults in the eight cities – Antarpur, Blasepur, Chastipur, Dustypur, Luckthen, Herebad, Matpur and Kantpur – are 50 + a, 70 + a, 30 + a, 80 + a, 30 + a, 40 + a, 50 + a and 20 + a, respectively.

Let the number of elderlies in Dustypur be e (since Dustypur has the highest total population of teenagers and adults, the population of elderlies in every other city will be more than e). Then the number of elderlies in Antarpur, Blasepur, Chastipur, Luckthen, Herebad, Matpur and Kantpur must be 30 + e, 10 + e, 50 + e, 50 + e, 40 + e, 30 + e and 60 + e, respectively.

The following table provides the populations of each section in each of the eight cities:

| City | Infants | Kids | Teenagers | Adults | Elderlies | Total |
|---|---|---|---|---|---|---|
| Antarpur | 10 | 20 | 40 | 10 + a | 30 + e | 110 + a + e |
| Blasepur | 20 | 10 | 50 | 20 + a | 10 + e | 110 + a + e |
| Chastipur | 40 | 20 | 20 | 10 + a | 50 + e | 140 + a + e |
| Dustypur | 10 | 10 | 60 | 20 + a | e | 100 + a + e |
| Luckthen | 30 | 30 | 30 | a | 50 + e | 140 + a + e |
| Herebad | 10 | 10 | 20 | 20 + a | 40 + e | 100 + a + e |
| Matpur | 30 | 20 | 40 | 10 + a | 30 + e | 130 + a + e |
| Kantpur | 40 | 30 | 20 | a | 60 + e | 150 + a + e |

**1.** Kantpur has the highest population.
Note that this question can be answered from a direct observation of the graph. Since it is given that the sum of Teenagers, Adults and Elderlies are the same across the eight cities, the city which has the highest total population of Infants and Kids will have the highest total population. From the graph, we can see that Kantpur has the highest total population of Infants and Kids and hence, will have the highest total population. Choice (B)

**2.** Number of adults in Antarpur = 10 + a
Number of adults in Dustypur = 20 + a
Given that $10 + a = \frac{2}{3}(20 + a) \Rightarrow a = 10$

Total number of adults (in mn) in all the eight cities combined = 90 + 8a = 170 Ans: (170)

**3.** Number of adults in Herebad = 20 + a
Number of elderlies in Chastipur = 50 + e
Given that $20 + a = 2(50 + e) \Rightarrow a = 80 + 2e$
Population of Matpur = 130 + a + e = 210 + 3e
Population of Kantpur = 150 + a + e = 230 + 3e
Required ratio = $\frac{210 + 3e}{230 + 3e}$
This value will be between 0.913 (for very low values of e) and 1 (for very high values of e). Choice (D)

**4.** Number of adults in Herebad = 20 + a
Number of elderlies in Kantpur = 60 + e
Given that these two values are equal.
Hence, a = 40 + e.
Also, the number of adults in Antarpur is the twice the number of elderlies in Blasepur.
10 + a = 2(10 + e) ⇒ a = 10 + 2e.
Hence, 10 + 2e = 40 + e ⇒ e = 30 and a = 70.
Population of Luckthen = 140 + 70 + 30 = 240. Choice (C)

**5.** For the number of adults to be the same as the number of elderlies in more than one city, the difference between the number of adults and the number of elderlies must be the same across these cities.
By observation, we can see that only for Antarpur, Herebad and Matpur, the number of elderlies is 20 more than that of adults. Hence, in these three cities, the number of adults must be the same as the number of elderlies.
Hence, a = 20 + e.
The ratio of the populations of Chastipur and Dustypur = $\frac{160 + 2e}{120 + 2e}$
This ratio lies between 1 and 1.333. Choice (A)

**Solutions for questions 6 to 10:**

Given that each student took at least three subjects. Hence, each student can take three or four or five subjects. Also, any student who takes OM also took OB and any student who took MAC also took FRA. Instead of drawing a five set Venn diagram, which is complicated, we can list down all the possibilities of subjects that the students can take.

The following table provides the list of subjects that the students can take, along with the variable used for representing the number of students who took that combination of subjects:

| Subjects | Variable |
|---|---|
| OM, OB, ECON | A |
| OM, OB, FRA | B |
| OB, ECON, FRA | C |
| OB, FRA, MAC | D |
| FRA, MAC, ECON | E |
| OB, ECON, FRA, MAC | F |
| OM, OB, FRA, MAC | G |
| OM, OB, ECON, FRA | H |
| All 5 subjects | I |

From (i), A + B + C + D + E = C + E + F + H + I ⇒ A + B + D = F + H + I ------ (1)
From (iii), F + G + H = 80 and **I = 100**. ------ (2)
From (iv), G + I = 130 ⇒ **G = 30**.
Hence, (2) can be written as F + H = 50.

Also, from (iv), E + F + I + 125 ⇒ E + F = 25 ----- (3)
Substituting the values of F + H and I in (1), we get A + B + D = 150.
From (v), the number of students who took MAC is D + E + F + G + I. We know that E + F is 25 and G is 30 and I = 100. Hence, the number of students who took MAC is 155 + D. From (v), **D = 0**

Also, from (v), B + C + D + F + H + G + I = 230 ⇒ B + C + 0 + 50 + 30 + 100 = 230 ⇒ B + C = 50.
Hence, B = 50 – C
A = 100 + C (since A + B + D = 150 ⇒ A + B = 150)
E = 25 – F and H = 50 – F.
From (ii), number of students who took OM = A + B + G + H + I = 150 + 30 + 50 – F + 100 = 330 – F.
Number of students who took ECON = A + C + E + F + H + I = 100 + C + C + 25 – F + F + 50 – F + 100 = 275 + 2C – F
Given that $\frac{330 - F}{275 + 2C - F} = \frac{9}{10} \Rightarrow 3300 - 10F = 2475 + 18C - 9F \Rightarrow 18C + F = 825$
$\Rightarrow C = \frac{825 - F}{18}$
For 825 – F to be a multiple of 18, F must be of the form 18k + 15.
Hence, F can be 15, 33, 51 and so on. However, E = 25 – F and hence, F cannot be greater than 25.
∴ **F = 15 and C = 45.**
**B = 5 and A = 145; E = 10 and H = 35.**

The following table provides the number of students for different combinations of subjects:

| Subjects | Number of Students |
|---|---|
| OM, OB, ECON | 145 |
| OM, OB, FRA | 5 |
| OB, ECON, FRA | 45 |
| OB, FRA, MAC | 0 |
| FRA, MAC, ECON | 10 |
| OB, ECON, FRA, MAC | 15 |
| OM, OB, FRA, MAC | 30 |
| OM, OB, ECON, FRA | 35 |
| All 5 subjects | 100 |

**6.** The total number of students in the college = 145 + 5 + 45 + 10 + 15 + 30 + 35 + 100 = 385. Ans: (385)

**7.** The number of students who took ECON but not OB = E = 10. Choice (A)

**8.** Number of students who took OB = 375
Number of students who took OM will be less than this (since all the students who took OM also took OB and also, 45 students took OB, ECON and FRA, who will not be counted here)
Number of students who took ECON = 350
Number of students who took FRA = 240
Number of students who took MAC will be less than this.
Hence, OB was taken by the maximum number of students. Choice (D)

**9.** The number of students who did not take only ECON will be the number of students who took all the subjects except ECON, i.e., G and G = 30. Hence, 30 students did not take only ECON. Choice (C)

**10.** Number of students who took exactly three subjects = A + B + C + D + E = 205
Number of students who took MAC = 155
Required difference = 50. Ans: (50)

**Solutions for questions 11 to 15:**

Given that every candidate knows the marks of all the other candidates. Also, each candidate can apply only once. Arranging the candidates in the descending order of their scores, we get:

| Candidate | Score | First Preference | Second Preference |
|---|---|---|---|
| E | 99.5 | IIB | IIS |
| F | 99.2 | IIB | IIK |
| B | 99.1 | IIS | IIK |
| L | 98.7 | IIS | IIB |
| O | 98.6 | IIB | IIK |
| A | 98.4 | IIS | IIB |
| D | 98.2 | IIK | IIB |
| I | 98.1 | IIS | IIB |
| M | 98.0 | IIK | IIS |
| G | 97.5 | IIS | IIK |
| J | 97.2 | IIB | IIS |
| N | 97.0 | IIB | IIS |
| K | 96.8 | IIB | IIK |
| C | 96.5 | IIB | IIS |
| H | 96.3 | IIK | IIB |

***In 2018:***
E will apply to the university of his first preference, i.e., IIB and he will be sure that he will be accepted. F also will apply to IIB, and will be sure that he will be accepted. B has IIS as his first preference. Even if E and F apply to IIS, B can still get admitted to IIS since it has three seats. Hence, B will also apply to IIS in 2018. L has IIS as his first preference. But he cannot be sure that he will get into IIS because if E, F and B apply to IIS, he will not be admitted. Hence, L will not apply in 2018. O has IIB as his first preference. If E, F, B and L apply to IIB, O can still apply because IIB has 6 seats. Hence, O will apply in 2018 and will be sure to get into IIB. A cannot apply because he cannot be sure if there will be vacant seats left in IIS if he applies. Similarly, every student below A cannot be sure. Hence, none of them will apply.

***In 2019:***
IIB has 3 vacancies, IIS has 2 vacancies and IIK has 6 vacancies at the end of 2018. Since L did not apply in 2018, the remaining candidates will know that L has IIS as his first preference (if he had any other university as his first preference, he would have applied in the previous year and would have been admitted). Similarly, every candidate will also know that A has IIS as his first preference following the same logic. L and A both will apply to IIS since there are two seats available. D knows that IIK will still have 6 vacancies after L and A apply. Hence, D will apply to IIK and can be sure that he will be selected. I knows that L and A will apply to IIS. Hence, I will apply to IIB and will be selected (since IIB has 3 vacancies and even if D applies to IIB, I can also get through). M will also apply to IIK since at most two people who have a better score than him can apply to IIK (I and D). G will also apply to IIK since he knows that IIS will be filled. J will not apply to IIB since more than 3 candidates would have applied to IIB (D, I, M and G all can apply, according to J). Similarly, N, K, C and H also will not apply.

***In 2020:***
IIB has 2 vacancies, IIS has 0 vacancies, and IIK has 3 vacancies. J will apply to IIB and N will also apply to IIB. Both will be admitted. K will know that J and N did not want to apply to IIK (if they did, they would have applied in 2019). Hence, K can deduce that J and N had either IIB as their first preference OR IIS as their first preference and IIB as their second preference. In either case, both will apply for IIB (since IIS is filled). K can deduce that there will be no seats remaining in IIB. Hence, K will apply to IIK, his second preference. C will also follow the same reasoning. Hence, C will not apply to IIB or IIS (since both will be filled). H will apply to IIK (since he knows that J and N could not have applied to IIK and even if C and K apply to IIK, there will be one more seat vacant).

The following table gives the university that each student joined every year:

| Year | IIB | IIS | IIK |
|---|---|---|---|
| 2018 | E, F, O | B | - |
| 2019 | I | L, A | D, M, G |
| 2020 | J, N | - | K, H |

**11.** The maximum number of students joined any university in 2019. Choice (B)

**12.** C did not join any university. Choice (C)

**13.** Three students, I, G and K, joined a university which was not their first preference. Choice (D)

**14.** One seat in IIK will be vacant in 2022. Choice (B)

**15.** H joined IIK in 2020. Choice (A)

**Solutions for questions 16 to 20:**

From (iii), B owns 5 mn shares in C2, C3 and C4. Since the percentage share should be a multiple of 5, the total number of shares of C2, C3 and C4 cannot be 80 mn or 120 mn. Hence, the number of shares of C2, C3 and C4 must be 20 mn or 50 mn or 100 mn.
From (ii), C owns 25% shares of C4. Let the number of shares of C4 that C owns be x. Then A must own x shares in C3.
If C4 has 20 mn shares, then x must be 5 mn. This is not possible as B already owns 5 mn shares of C4 (with percentage share of 25%).
If C4 has 50 mn shares, then x must be 12.5 mn. But this is not possible as A cannot own 12.5 mn shares of C3 as it will not result in a multiple of 5 irrespective of whether C3 has 20 mn shares or 100 mn shares. Hence, this is also not possible.
Hence, C4 must have 100 mn shares. The value of x must be 25 mn. Since A owns 25 mn shares of C3, C3 cannot have 20 mn shares. C3 must have 50 mn shares. C2 will have 20 mn shares.

From (iii), the percentage share of D in C4 must be 10% (since percentage share of B in C3 is 10%). Similarly, the percentage share of D in C5 must be 5%.
The number of shares of C4 that A owns must be 100 – 5 – 25 – 10 = 60 mn. From (i), A must own 60 mn shares of C1.

C1 can have 80 mn or 120 mn shares. If C1 has 80 mn shares, then A will have 75% share (with 60 mn shares). The others should have a minimum 5%, 10%, 15% share in C1. However, this is not possible as the sum of the percentage shares will exceed 100.
Hence, C1 must have 120 mn shares and A will have 50% share in C1.
C5 must have 80 mn shares and D owns 5% of it, i.e., 4 mn shares of C5.

From (v), A owns equal number of shares in C2 and C5. In C2, A can own any number of shares from 1 to 12 mn (since B owns 5 mn shares and the rest, C and D, must own at least 1 mn and 2 mn shares). In C5, A can own 4/8/12/16 mn shares. For these two to be equal, A must own either 4 or 8 or 12 mn shares, in each of C2 and C5. But A cannot own 4 mn shares in C5 since D owns 4 mn shares in C5. Also, if A owns 12 mn shares in C2, then C and D will own 1 mn and 2 mn shares. From (iv), the number of shares that D can own in C1 will be 1 mn or 2 mn. This is not possible because the percentage share of D in C1 will not be a multiple of 5 in this case. Hence, A cannot own 12 mn shares in C2. Hence, A must own 8 mn shares in each of C2 and C5.
The number of shares that C owns in C5 must be a multiple of 4 mn. The number of shares of C1 that B owns must be a multiple of 6 mn (only then will the percentage shares of two values correspond to multiples of 5). For these two to be equal, the only possibilities are 12/24/36/48/60.
Of these 60 is not possible since for C1, A owns 60 mn shares out of 120 mn. B cannot own the remaining 60 mn. Also, 48 mn is not possible because if B owns 48 mn shares, then A and B will own 108 mn shares. The others must own 12 mn shares and must own 5% each. In this case, the percentage shares will not be distinct. Hence, the only possibilities are 12/24/36.
Since C owns 12/24/36 mn shares in C5, A owns 8 mn shares and D owns 4 mn shares, B can own 56/44/32 mn shares in C5. For (v) to be satisfied, B must own 36 mn shares in C1 and 32 mn shares in C5.
In C2, C can own any number of shares from 1 to 6. In C1, D can own 6/12/18... mn shares. From (iv), these two must be equal. This is possible only if C and D own 6mn shares in C2 and C1, respectively.
C must own 18 mn shares in C1 and D must own 1 mn shares in C2. In C3, C and D can own (2.5, 17.5) or (7.5, 12.5) mn shares, in any order.

The following table provides the number of shares that each person has in each of the five companies:

| | C1 | C2 | C3 | C4 | C5 |
|---|---|---|---|---|---|
| A | 60<br>*(50%)* | 8<br>*(40%)* | 25<br>*(50%)* | 60<br>*(60%)* | 8<br>*(10%)* |
| B | 36<br>*(30%)* | 5<br>*(25%)* | 5<br>*(10%)* | 5<br>*(5%)* | 32<br>*(40%)* |
| C | 18<br>*(15%)* | 6<br>*(30%)* | | 25<br>*(25%)* | 36<br>*(45%)* |
| D | 6<br>*(5%)* | 1<br>*(5%)* | | 10<br>*(10%)* | 4<br>*(5%)* |
| Total | 120 | 20 | 50 | 100 | 80 |

**16.** B owns 36 mn shares of C1. Ans: (36)

**17.** The difference (in mn) between the number of shares of C2 that A owns and the number of shares of C1 that C owns = 18 – 8 = 10 mn. Ans: (10)

**18.** C owns the maximum number of shares of C5. Choice (B)

**19.** The number of shares of C3 that D owns can be 2.5/7.5/12.5/17.5. This is less than the number of shares of C1 that C owns. Choice (C)

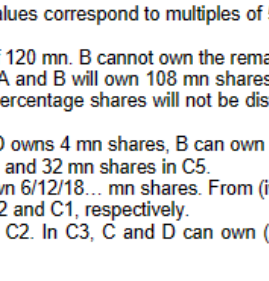
**20.** The total number of shares that A owns = 161 mn.
The total number of shares that B owns = 83 mn.
Required difference = 78 mn. Ans: (78)

| *Difficulty level wise summary - Section II* | |
|---|---|
| Level of Difficulty | Questions |
| *Very Easy* | – |
| *Easy* | 1, 2, 4 |
| *Medium* | 3 |
| *Difficult* | 5, 6, 7, 8, 9, 10, 13, 14, 16, 17, 18, 19, 20 |
| *Very Difficult* | 11, 12, 15 |

### SECTION – III

**Solutions for questions 1 to 22:**

**1.**



Let the larger and the smaller perpendicular sides of the rectangle be a, b. Let the hypotenuse be c.
$(a + b) - c = a/3$ and $c = \sqrt{a^2 + b^2}$
$a + b - \sqrt{a^2 + b^2} = a/3$
$\Rightarrow (2/3)a + b = \sqrt{a^2 + b^2}$
squaring on both sides $= \frac{4a^2}{9} + \frac{4ab}{3} + b^2 = a^2 + b^2$
$\Rightarrow 12ab = 5a^2 \Rightarrow \frac{b}{a} = \frac{5}{12}$
Hence the ratio of the lengths of the shorter side to the longer side = 5 : 12

**Alternative Solution:**

The equation can be solved using the options.
(A) 8 : 15
Total length along sides is (8 + 15) = 23, and along the diagonal is 17.
Difference $= 6 \neq \frac{1}{3}\times 15 \left(\frac{1}{3} \text{ of longer side}\right)$
(B) 3 : 4
Length along sides = (3 + 4) = 7
Diagonal = 5
Difference $= 2 \neq \frac{1}{3}\times(4)$
(C) 7 : 24
$(24 + 7) - 25 = 6 \neq \frac{1}{3}\times 24$
(D) 5 : 12
$(5 + 12) - 13 = 4 = \frac{1}{3}\times 12$ Choice (D)

**2.** Given that $36a^2 + \frac{1}{9a^2} = 4$
$(6a)^2 + \left(\frac{1}{(3a)}\right)^2 = 4$
$(6a)^2 + \left(\frac{1}{(3a)}\right)^2 - 2.6a\frac{1}{(3a)} = 4 - 2.6a.\frac{1}{(3a)}$
$\left(6a - \frac{1}{3a}\right)^2 = 4 - 4 = 0$
$\left(6a - \frac{1}{3a}\right) = 0$, i.e., $6a = \frac{1}{(3a)}$
$(6a)^3 - \left(\frac{1}{(3a)}\right)^3 = 0$
$216a^3 - \frac{1}{27a^3} = 0$ Ans: (0)

**3.** From Wilson's theorem, (29 – 1)! + 1 is divisible by 29.
From Fermat's little theorem, $(27^{(29-1)} - 1)$ is divisible by 29.
Hence, $(29-1)! + 27^{28}$ is divisible by 29. Therefore, the required remainder will be 0. Choice (A)

**4.** The indices of the terms in the two GPs are in AP.
$G_1 : 2^2, 2^{3.5}, 2^5, \ldots\ldots 2^{269}$
$G_2 : 2^3, 2^{4.75}, 2^{6.5}, \ldots\ldots 2^{269}$
The two APs are
$A_1$ : 2, 3.5, 5, ......269
$A_2$ : 3, 4.75, 6.5, ......269
We can get rid of fractions by considering
$4A_1$ : 8, 14, 20, 26, ....... 1076
$4A_2$ : 12, 19, 26, ......... 1076
The common differences are 6 and 7 (LCM = 42) respectively.
Therefore, the AP of the common terms is
A : 26, 68, 110, ..... 1076
Number of terms $= \frac{1076 - 26}{42} + 1 = 26$ Ans: (26)

**5.** We have to evaluate $A_i$ for successive values of i, for which we need to evaluate $f(A_{i-1})$. The first value of n (i.e., 0), the corresponding value of A (i.e., $A_0$) the subsequent values of i and the corresponding values of $A_i$ are tabulated below.

| i | $f(A_{i-1})$ | $A_i$ |
|---|---|---|
| 0 | – | 13 |
| 1 | 26 | 52 |
| 2 | 49 | 98 |
| 3 | 95 | 190 |
| 4 | 187 | 374 |
| 5 | 371 | 742 |
| 6 | 739 | 1478 |

∴ $A_6$ = 1478 Choice (B)

**6.** Let I be the point of intersection of the diagonals. The diagonals are perpendicular to each other.
$\therefore \angle PIS = \angle PIQ = \angle SIR = \angle RIQ = 90^\circ$
$PS^2 + QR^2 = PI^2 + IS^2 + IQ^2 + IR^2$
$\Rightarrow (PI^2 + IQ^2) + (IS^2 + IR^2) = PQ^2 + RS^2$
$PS^2 + 12^2 = 16^2 + 20^2$
$\Rightarrow PS = 16\sqrt{2}$ Choice (C)

**7.** $\frac{1}{12} + \frac{1}{20} + \frac{1}{30} + \frac{1}{42} + \ldots\ldots$
$= \frac{1}{3(4)} + \frac{1}{4(5)} + \frac{1}{5(6)} + \frac{1}{6(7)} + \ldots\ldots \infty$
$t_n = \frac{1}{(n+2)(n+3)} = \frac{1}{n+2} - \frac{1}{n+3}$
$t_1 = \frac{1}{3} - \frac{1}{4}$
$t_2 = \frac{1}{4} - \frac{1}{5}$
$t_3 = \frac{1}{5} - \frac{1}{6}$
$\therefore t_n = \frac{1}{n+2} - \frac{1}{n+3}$
$S_n = t_1 + t_2 + t_3 + \ldots\ldots t_n$
$= \left(\frac{1}{3} - \frac{1}{4}\right) + \left(\frac{1}{4} - \frac{1}{5}\right) + \left(\frac{1}{5} - \frac{1}{6}\right) + \ldots\ldots + \left(\frac{1}{n+2} - \frac{1}{n+3}\right)$

$S_n = \frac{1}{3} - \frac{1}{n+3}$
As $n \to \infty$ $\frac{1}{n+3} \to 0$ and $S_n \to \frac{1}{3}$
$\therefore S = \frac{1}{12} + \frac{1}{20} + \frac{1}{30} + \frac{1}{42} + \ldots = \frac{1}{3}$
Hence, 72S = 24 Ans: (24)

**8.** The minute hand covers 360° in one hour, whereas the hour hand covers 30° in one hour. Hence, minute hand moves at a relative (angular) speed of $\frac{360^\circ}{60} - \frac{30^\circ}{60} = 5.5^\circ$ per minute faster than the hour hand. (This result may be committed to memory for convenience).
Given the angle between the minute hand and hour hand was 55° when Harshita left and also when she returned.
Now, the minute hand could have been 55° behind the hour hand when she left and then travelled (overtaken) to exactly 55° ahead of it by the time she returned. In which case, the time elapsed would be $\frac{55^\circ + 55^\circ}{5.5^\circ} = 20$ min.
However, it is possible that the minutes hand was already 55° ahead (when she left) and then moved through 360° – (55° + 55°), i.e., 250°, to arrive at a position that was 55° behind the hour hand (when she returned). The time elapsed for this case would be $\frac{250^\circ}{5.5^\circ} = 45.45$ min.
Any other possible configuration of initial and final positions would take more time than 45.45 min. For example, the minute hand could move through exactly 360°, (i.e., from being 55° behind to again being exactly 55° behind) taking $\frac{360}{5.5} = 65.45$ minutes.
Since it is given that she returned in less than 45 minutes, she must have spent 20 minutes outside. Choice (D)

**9.** The only value of $\sqrt{x}$ for which this equation will be valid is when $\sqrt{x} = 2$, since $3^2 + 4^2 = 5^2$.
Hence, $\sqrt{x} = 2$. Therefore x = 4. Choice (B)

**Note:** For any real number, a, $3^a + 4^a > 5^a$, for a < 2 and $3^a + 4^a < 5^a$, for a > 2.

**10.** If N is the number of rows, then the number of students is $N^2$.
Let k be the number of packets used (this includes the one packet in which a few chocolates were remaining).
The total number of chocolates in these k packets = 240k
If the number of chocolates left in the last packet was 10a, then the number of chocolates distributed is 240k – 10a.
Since each student was given 10 chocolates, the number of students must be $\frac{240k - 10a}{10} = 24k - a$, which must be equal to $N^2$, which is the square of an odd number, which must itself be odd.
Since 24k is always even, a must be odd, for 24k – a to be an odd number.
The options provide different values for 10a but only for option B, the value of a is an odd number. Hence, 150 chocolates must have been left in the packet. The answer must be option B.

**Note:** In case there were more than one odd number for the value of a among the choices, then the value of 24k – a can be calculated for all these options, using the approach given below. Any choice which yields a feasible value of k must be the correct answer.
For example, we can calculate the values of 24k – 15 for different values of k. Trying k = 1, 2, 3..., we get 24k – 15 = 9 for k = 1, which is an odd square.

**Alternative solution:**

The number of students in the school = (Number of rows) × (Number of students per row) = (Number of rows)² (∵ Number of rows = Number of students per row).
∴ Number of students is a perfect square.
Let this be $N^2$ (where N is odd). Let the number of packets from which chocolates were distributed be T. Total number of chocolates distributed = $10N^2$ (because each student got exactly 10 chocolates).
Let the remaining number of chocolates in the last packet be R.
Then $240T = 10N^2 + R$ – (1)
$\Rightarrow N^2 = 24T - \frac{R}{10}$
$\therefore \frac{-R}{10}$ denotes the remainder when 24 divides $N^2$, which is a perfect square. Any odd perfect square when divided by 24 must leave a remainder of 1 or 9.
$\therefore \frac{-R}{10} = 1$ or 9
$\therefore \frac{R}{10} = -1$ or $-9$
$\therefore \frac{R}{10} = (-1 + 24)$ or $(-9 + 24) = 23$ or 15
i.e., R = 230 or 150.
∴ From the given choices, only Choice (B) can be a possible value of R. Choice (B)

**11.** Points which are equidistant from the coordinate axes must lie on the line y = x or y = –x.
Therefore, points on the line 2x + 3y = 6 which are equidistant from the coordinate axes must be those points where 2x + 3y = 6 intersects y = x or y = –x.
Points of intersection of 2x + 3y = 6 and y = x is $\left(\frac{6}{5}, \frac{6}{5}\right)$, whereas the lines 2x + 3y = 6 and y = –x intersects at (–6, 6).
Therefore, two points on the given line are equidistant from the coordinate axes. Choice (C)

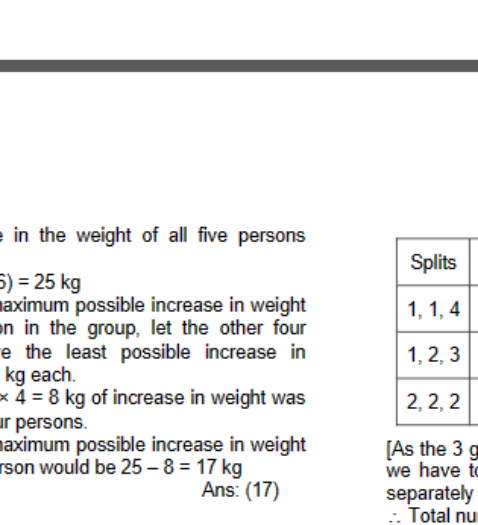
**12.** The increase in the weight of all five persons together
= 5 × (81 – 76) = 25 kg
To find the maximum possible increase in weight of any person in the group, let the other four persons have the least possible increase in weight, i.e., 2 kg each.
Therefore, 2 × 4 = 8 kg of increase in weight was due to the four persons.
Hence, the maximum possible increase in weight of the fifth person would be 25 – 8 = 17 kg Ans: (17)

**13.** The different splits and the number of such splits of the 6 people into 3 groups are tabulated below.

| Splits | Selecting vehicles | Selecting persons | |
|---|---|---|---|
| 1, 1, 4 | 3 | $^6C_4\ ^2C_1$ | 15(2) (3) = 90 |
| 1, 2, 3 | 6 | $^6C_3\ ^3C_2$ | 20(3) (6) = 360 |
| 2, 2, 2 | 1 | $^6C_2\ ^4C_2$ | 15(6) (1) = 90 |

[As the 3 groups have to go in 3 distinct vehicles, we have to count A, B, CDEF and B, A, CDEF separately etc]
∴ Total number of ways = 90 + 360 + 90 = 540 Ans: (540)

**14.**



The two curves have been plotted above. Their points of intersection is (0, 0) and (2, 0). The area enclosed between the two curves is the shaded area shown above $= 2\left[\frac{1}{2}\times 2\times 1\right] = 2$ sq. units. Ans: (2)

**15.** Let the measure of the third side be x, where x is a natural number.
From the triangle inequality, sum of any two sides must be greater than the third side.
The three sides are 6, 10 and x
We have two cases:
Case I: x is the longest side
∴ 6 + 10 > x ⇒ x < 16
Case II: 10 is the longest side
∴ 6 + x > 10 ⇒ x > 4
Therefore, the possible values of x are 4 < x < 16, i.e., there are 11 possible values for the third side (x = 5, 6, 7....14, 15). Ans: (11)

**16.** 2a = 3b = 5c ------ (1)
Multiplying by 4, we get 8a = 12b = 20c
Similarly 4c = 5d = 6e ------ (2)
Multiplying by 5, we get 20c = 25d = 30e
∴ 8a = 12b = 20c = 25d = 30e = N (say)
Now, $b = \frac{N}{12}, c = \frac{N}{20}, a = \frac{N}{8}$ and $e = \frac{N}{30}$
$\therefore \frac{bcd}{ae} = \frac{\left(\frac{N}{12}\right)\left(\frac{N}{20}\right)d}{\left(\frac{N}{8}\right)\left(\frac{N}{30}\right)} = d$ Choice (D)

**17.** Let x be the number of boys who participated and let y be the number of points scored by each of them.
Hence the total number of points won in the tournament is xy + 8.
Clearly this number is equal to the total number of games played.
Since the total number of players is (x + 2) and each of them played with all the others once, the total number of games is
$^{x+2}C_2 = \frac{(x+2)(x+1)}{2}$
$\therefore xy + 8 = \frac{(x+2)(x+1)}{2}$
$\Rightarrow 2xy + 16 = x^2 + 3x + 2 \Rightarrow x^2 + 3x - 2xy = 14$
$\Rightarrow x(x + 3 - 2y) = 14$
Clearly x is an integer.
Since y is either an integer or a fraction, with denominator 2,
(x + 3 – 2y) is also an integer.
Since x must divide 14, x can be 1, 2, 7 or 14.
But x = 1 and x = 2 is not possible since in either of these cases the total number of students will not exceed 4 and two girls cannot score 8 points.
Hence x = 7 or 14
If x = 7
7(7 + 3 – (2)(4)) = 14
⇒ y = 4

If x = 14
14(14 + 3 – (2)(8)) = 14
⇒ y = 8
Hence the number of boys in the tournament can be either 7 or 14.

**Alternate method:**

The answer can be verified from the options,
$xy + 8 = \frac{(x+2)(x+1)}{2}$
$x = 7 \Rightarrow 7y + 8 = \frac{9\times 8}{2} = 36$
⇒ y = 4
∴ x = 7 satisfies
$x = 11 \Rightarrow 11y + 8 = \frac{13\times 12}{2} = 78 \Rightarrow y = \frac{70}{11}$
This is not a valid value of y since y can take only multiples of $\frac{1}{2}$.
$x = 14 \Rightarrow 14y + 8 = \frac{16\times 15}{2} = 120 \Rightarrow y = 8$
This is valid. ∴ x = 7 or 14 satisfies. Choice (A)

**18.** If $3x^2 + ax - 3x + 12 = 0$ has real and equal roots,
$(a - 3)^2 - 4(3)(12) = 0$
⇒ a – 3 = ± 12
∴ a = 15 or a = –9.
Therefore, the difference between the maximum and minimum possible values of a = 15 – (–9)
= 24 Choice (D)

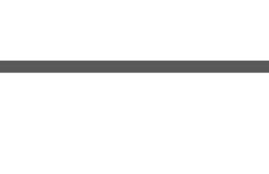
**19.** The planes land after 0, 35, 70, 105... (35×n) minutes from 8:00 AM.
The planes take off after 5, 25, 45, 65... (20k + 5) minutes from 8:00 AM.
The instances when a landing and a take-off can occur at the same time will be first at 105 (35 × 3, 20 × 5 + 5) and then, after every 140 minutes.
The number of minutes between 8:00 AM and 9:30PM
= 810 minutes.
Hence, the planes will get delayed at 105 min, 245 min, 385 min, 525 min, 665 min and 805 min.
Hence, a total of 6 planes will get delayed. Choice (D)

**20.** $x = \cfrac{1}{4 - \cfrac{1}{4 + \cfrac{1}{4 - \cfrac{1}{4 + \ldots\infty}}}} \Rightarrow x = \cfrac{1}{4 - \left(\cfrac{1}{4 + x}\right)}$
$x = \frac{4 + x}{(16 + 4x - 1)} \Rightarrow 15x + 4x^2 = 4 + x$
$\Rightarrow 4x^2 + 14x - 4 = 0$
$\Rightarrow 2x^2 + 7x - 2 = 0$
$\therefore x = \frac{-7 \pm \sqrt{49 + 16}}{4} = \frac{-7 \pm \sqrt{65}}{4}$
As x cannot be a negative quantity, $x = \frac{\sqrt{65} - 7}{4}$

**Alternative Solution:**

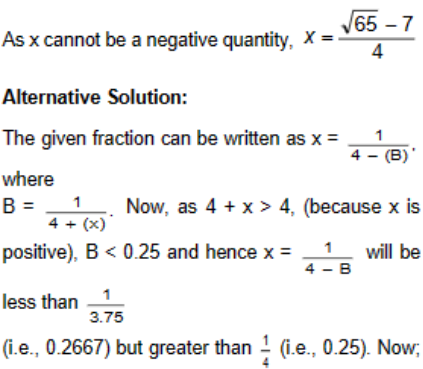
The given fraction can be written as $x = \frac{1}{4 - (B)}$, where $B = \frac{1}{4 + (x)}$. Now, as 4 + x > 4, (because x is positive), B < 0.25 and hence $x = \frac{1}{4 - B}$ will be less than $\frac{1}{3.75}$ (i.e., 0.2667) but greater than $\frac{1}{4}$ (i.e., 0.25). Now; as 0.25 < x < 0.2667 using the onscreen calculator we can check for each of the given options. Clearly, choices (B) and (D) are negative and choice (C) ≅ 0.292. Hence, only choice (A) satisfies. Choice (A)

**21.** Let the number of ways in which one can go from A to B, B to C and A to C directly be denoted by a, b and c respectively.



It is given that, a = c = 8 and
∴ ab + c = 32
⇒ 8b + 8 = 32
⇒ b = 3
⇒ Therefore, one can go from B to C in 3 ways. Choice (B)

**22.**



Consider the given diagram
Now, X∆Y = {a, f, b, e}
∴ (X∆Y) ∆Z = {a, f, b, e} ∆ {e, f, g, c}
X∆Y∆Z = {a, b, c, g}
∴ X∆Y∆Z is a set of elements which either belong to exactly one of the sets or to all the three sets.
In the interval 1 to 100,

The number multiples of only 2 = 27
The number multiples of only 3 = 14
The number multiples of only 5 = 7
The number of multiples of 2, 3, and 5 = 3
∴ required result = 27 + 14 + 7 + 3 = 51 Ans: (51)

| *Difficulty level wise summary - Section III* | |
|---|---|
| Level of Difficulty | Questions |
| *Very Easy* | – |
| *Easy* | 12, 16, 18 |
| *Medium* | 1, 2, 4, 5, 6, 8, 9, 11, 13, 15, 19, 20, 21, 22 |
| *Difficult* | 3, 7, 14, 17 |
| *Very Difficult* | 10 |

Done
Tip: Scroll slowly for the best results